



Figure 8: R Studio Environment and console

```
> a-b
[1] -20 -20 -20 -20 -20
> a*b
[1] 21 44 69 96 125
```

A list is like a vector, but can store arbitrary or any type of data. To create a list, the 'list' function is used, as follows:

```
> l <- list(1,2,3,"ABC")
> l
[[1]]
[1] 1

[[2]]
[1] 2

[[3]]
[1] 3

[[4]]
[1] "ABC"
```

A list can be used to hold different types of objects. It can be used to store a vector, list, data frame or anything else.

An array is nothing but a multi-dimensional vector that can store data in rows and columns. An array function is used to create an array.

```
> arr <- array(21:24, dim=c(2,2))
> arr
      [,1] [,2]
[1,] 21  23
[2,] 22  24
```

A data frame and matrix are used to hold tabular data. It can be thought of as an Excel sheet with rows and columns. The only difference between a data frame and matrix is that in the latter, every element should be of the same type. The following code shows how to create the data frame:

```
> x<-1:5
> y<-c("ABC", "DEF", "GHI", "JKL", "MNO")
> z<-c(25,65,33,77,11)
> d <- data.frame(SrNo=x, Name=y, Percentage=z)
> d
```

	SrNo	Name	Percentage
1	1	ABC	25
2	2	DEF	65
3	3	GHI	33
4	4	JKL	77
5	5	MNO	11

So a data frame is nothing but a vector combined in the column format.

This article gives a basic idea of how data is handled by R. I leave the rest for you to explore. 

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HyperLedger: Hyperledger nurtures and endorses a wide array of businesses around blockchain technologies, including distributed ledgers, smart contracts, etc. Hyperledger encourages the re-use of common building blocks and enables the speedy invention of distributed ledger technology components.

Project link: <https://hyperledger.org/projects>

Project GitHub link: <https://github.com/hyperledger>

Openchain: Openchain is an open source distributed ledger technology. It is ideal for enterprises, and deals in issuing and managing digital assets in a robust, secure and scalable way.

Project link: <https://www.openchain.org/>

Ethereum project: This is a distributed framework that runs smart contracts—applications that run exactly as programmed in a secured virtual environment without

downtime or the possibility of tampering, as this platform leverages the custom built blockchain.

Project link: <https://www.ethereum.org/>

There are some more blockchain projects, links to which can be found in the *References* section. 

Reference

[1] <https://lightrains.com/blogs/opensource-blockchain-platforms>

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